SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5109

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I K.Mounith(192425090)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-bystep.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEEFEAOCAIVDEN”.

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation n with missing details	Poor or incomplete documentation

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1) Scenario:- event communication security
A student organization preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into a encoded version using a fixed shift team pattern.

a. Re construct how the encrypted message midnight appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of encoding rule, would retrive original message.

2) Scenario: Intercepted suspicious message A cyber security intern intercepts the coded message: "WS RHI RH ZHIVXIVW" during network monitoring. The structure of text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce to most likely original message.

b. Justify your answer by analyzing repeating patterns, letter structures or known cipher characteristics.

Scenario:- Secure message system.
A company deploys a keyword-based encryption

technique for internal communication. An employee sends the word "HELLO" using keyword "KEY" to ensure confidentially.

a. Demonstrate how encryption process transforms message step-by-step.

b. explain how Repeating keyword influences each character of ciphertext.

4) Scenario: Legacy encryption in corporate communication A Legacy in a company still uses an transposition-based encryption method with three levels of arrangment. A recieved message appears as: "WECRCTEERDSOEEFTEAOCAIVDEN"

a. determine most probable original message by reversing encryption process.

b. critically evaluate why such a method is vulnurable in modern communication Systems.

c. propose one pratical enchancement to Improve the confidentially of message in this Scenario.

1) given key = 4

word = "MEET AT NOON"

It is under "Substitution method" and it is also known as "SHIFT Cipher".

a) Shifting of H :-

word = MEET AT NOON | Key = 4

formula = (P+K) mod 26

we need to shift the word by word 4 digits

M = (12+4) mod 26 = 16 mod 26 = 16 = Q

E = (4+4) mod 26 = 8 mod 26 = 8 = I

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T = (19+4) mod 26 = 23 mod 26 = 23 = X

A = (0+4) mod 26

T = X

N = (13+4) mod 26 = 17 = R

O = (14+4) mod 26 = 18 = S

O = S

N = R

MEET AT NOON = "QIIX EX RSSR"

Encrypted value.

b) decryption : $c = (k-p) \bmod 26$ [key = 4]

encrypted for value

decrypted word: "Φ I I X E X R S S R"

$\Phi = ((6-4) \bmod 26 = 12 \Rightarrow m$

$I = (8-4) \bmod 26 = 4 \Rightarrow E$

$I = (8-4) \bmod 26 = 4 \Rightarrow E$

$X = (23-4) \bmod 26 = 19 \Rightarrow T$

$E = (4-4) \bmod 26 = 0 \Rightarrow A$

$X = (23-4) \bmod 26 = 19 \Rightarrow T$

$R = (17-4) \bmod 26 = 13 \Rightarrow N$

$S = (18-4) \bmod 26 = 14 \Rightarrow O$

$S = (18-4) \bmod 26 = 14 \Rightarrow O$

$R = (17-4) \bmod 26 = 13 \Rightarrow N$

decryption word

"MEET AT NOON"

∴ cipher text to plain text

2) given code message is "ᗡSR H RH ZHVXIVᗡ"

a) A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Z Y X W V U T S R Q P O N M L K J I H G F E D C B A.

now word = GSRH RHZ HVXIVG

THIS IS A SECRET

4) given word = WJRH RHZ HVXIVH

This word is divided into "4 2 1 6" That matchg the word called this is a secret and for this we can't Apply only mono arthematic shift or cipher shift. Because If we are Applying any method we are getting words.

3) a) given word : "HELLO"

" key : "key"

" This is vigenere cipher method"

encryption: we need to add given word with key word and If it is small repeate addit.

$$C = (P + K) \bmod 26$$

H	E	L	L	O
K	E	Y	K	E
R	I	J	V	S

encryption word is

" R I J V S"

b) ciphertext. quick and fast

eg:- H E C C o "Here o is key = H"
H H H H H

here down value is always 1 so, no need to calculate every time. So this like the repeating unit should be like.

4) a) given word = "W E C R C T E E R D S A E E T E A O C A 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
I V D E W". This under Transposition "Rail fence."

Depth = 3

here total characters = 25/3 = 8 Rem -1

Rail 1 Totally we are having 3 Rails
Rail 2 here now we are going to decide
Rail 3 The which Rails need to put which letter.

Rail 1: W E C E T C
Rail 2:- E R O S O V D N
Rail 3: A I R A O E

word: "WE ARE DISCOVERED FLEE AT ONCE"